Narrowband OTA Phase Calibration

Multipath Statistics, Spatial Averaging and Sampling

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26 runs × 5 carriers
Indoor upper mid-band measurement campaign

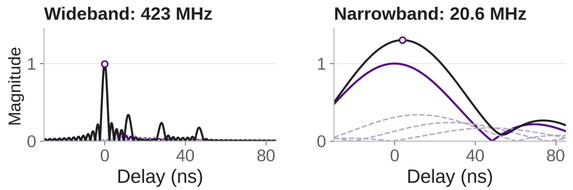
30 / 30
Pooled CDF fits improve over Gaussian baseline

29.5% lower median RMSE
Random → systematic · 598 mm · 5% BW · 50 obs.

128 / 130
Configurations improved · 5% BW · 50 samples

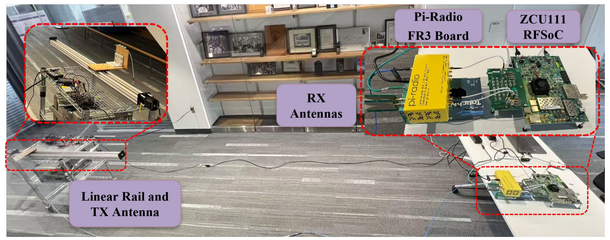
1 PROBLEM & MEASUREMENTS

LOS phase supports array calibration, beamforming, localization and sensing. Limited bandwidth leaves reflections unresolved, perturbing the observed phase.



Synthetic example: purple = LOS; black = complex sum.
A dominant delay peak can still contain phase error.

Indoor measurement campaign



26 runs: labs, meeting rooms, cubicles and hallways.
Radial TX motion preserves a nearly constant LOS angle.

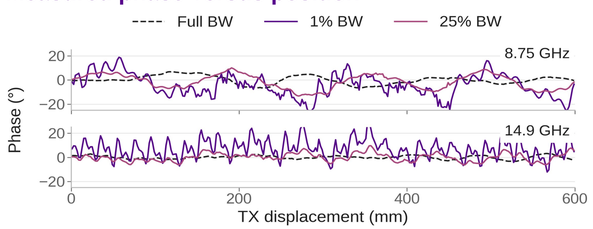
Hardware	ZCU111 RFSoC + coherent Pi-Radio
Carriers (GHz)	8.75, 10.25, 12.975, 13.9, 14.9
Spatial coverage	300 positions; 2 mm steps; 598 mm span
Per position	15 frames × 5 symbols = 75 observations

984 active subcarriers × 430 kHz = 423.12 MHz.
Pre/post chain change: 2.16 dB and 5.13° max. in one example.

Paired measurement processing

1 Synchronize and estimate CFR → 2 Align channels; suppress leakage → 3 Compensate hardware response → 4 Apply NB / WB masks to both RXs → 5 64× zero pad; IFFT and peak → 6 Form paired channel ratios

Measured phase versus position



Example run: RX1 phase relative to the RX2-WB reference.
Narrower bands reveal structured spatial fluctuations.

2 STATISTICS & SPATIAL AVERAGING

Normalize the LOS component to one. A weaker aggregate residual perturbs the measured phase:

$$h = 1 + \beta e^{j\phi}, \quad \theta = \arg(h), \quad 0 \le \beta < 1$$

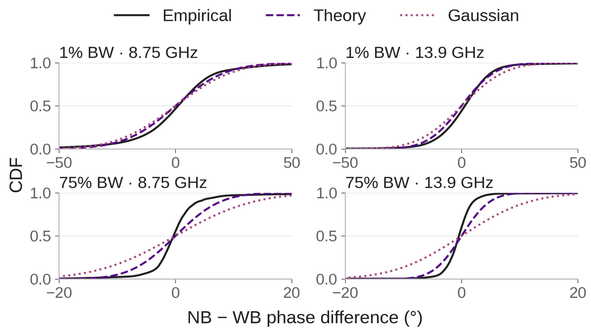
Residual phase is uniform and amplitude-independent; then phase error is symmetric and has zero mean.

$$\widetilde{f}_\theta(\theta) = \frac{\cos\theta}{\sqrt{2\pi\,\zeta C_\zeta}}\, e^{-\sin^2\theta/(2\zeta^2)}$$

$$\widetilde{F}_\theta(\theta) = \frac{\Phi(\sin\theta/\zeta) - \Phi(-1/\zeta)}{C_\zeta}$$

$C\zeta = 2\Phi(1/\zeta) - 1$; interior formulas for $|\theta| < \pi/2$.
Rayleigh amplitude scale ζ; normalized weak-residual model.

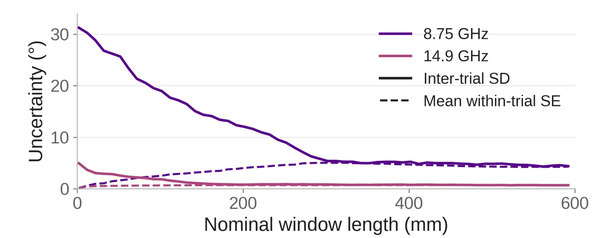
Measured distribution validation



30 / 30 pooled bandwidth–frequency settings

Smaller ECDF discrepancy than the RMS-fitted Gaussian.
All samples retained; this is a descriptive pooled-data fit.

Spatial correlation limits averaging



$$\sigma_{\text{inter}} = \mathrm{SD}_t(\bar{e}^{(t)}), \quad \sigma_{\text{intra}} = \mathrm{mean}_t(s^{(t)}/\sqrt{50})$$

Lab 2, side reflector; 5% BW; 50 observations per trial.
At 8.75 GHz, inter-trial SD falls from 31.3° to 4.4°.
Agreement of the two curves indicates stability; it does not establish independence or zero residual bias.
Weak-perturbation limit: the normalized model becomes Gaussian.

3 NARROWBAND CALIBRATION

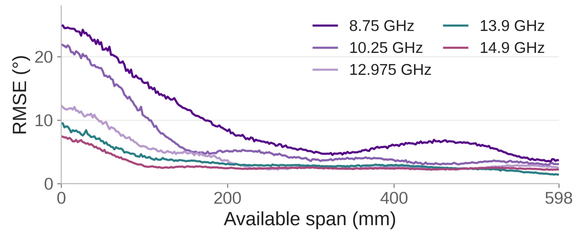
Estimate relative phase using two narrowband channels. Normalize each ratio to give every phase equal weight.

$$q_n = \frac{h^1_{\text{NB},n}}{h^2_{\text{NB},n}}, \quad \hat{\theta} = \arg\left(\sum_{n\in\mathcal{A}} \frac{q_n}{|q_n|}\right)$$

$$\hat{c}_1 = e^{-j\hat{\theta}}, \quad \hat{c}_2 = 1; \qquad \mathrm{RMSE}^2 = b^2 + \sigma^2$$

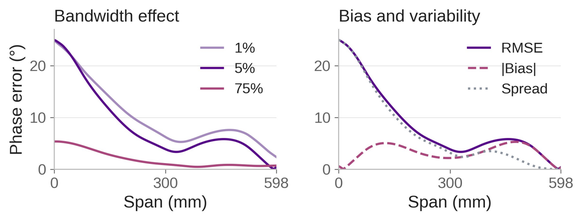
All-data estimator: 75 observations per position.
Fixed-budget estimator: 50 observations per window.
WB reference: all 22,500 observations over 598 mm.

Error versus spatial coverage



Illustrative run · 5% bandwidth · fixed 50 observations.
Spatial coverage helps at the same observation count.

Bandwidth, bias and variability



Same run, 8.75 GHz; all observations. Wider bandwidth reduces short-window error; final offsets need not vanish.

Calibration endpoints at 5% bandwidth

Carrier (GHz)	One position 75 obs.	598 mm 22,500 obs.	598 mm 50 obs.
8.750	24.98	0.38	3.69
10.250	21.57	0.24	3.10
12.975	12.34	1.77	2.46
13.900	9.34	0.58	1.45
14.900	7.49	1.97	2.24

Entries in degrees: single-position RMSE / full-trajectory absolute error / fixed-budget RMSE, respectively.

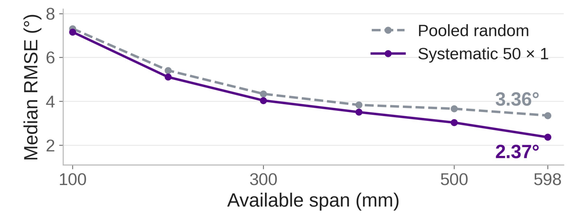
WB agreement is measured; absolute hardware phase and complete angular calibration are not identified.

4 SAMPLING & RELIABILITY

Use the same estimator and 50-observation budget.
Change only how the spatial positions are selected.

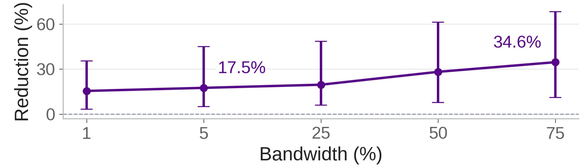


Random: sample from the window's 75w observations.
Systematic: one sample at each of 50 spaced positions.



5% BW · 130 configurations · fixed 50 observations.
At 598 mm: 3.36° → 2.37° median RMSE (−29.5%).

Paired gain across bandwidths



Median and IQR at 598 mm across 130 configurations.
At 5% BW: median paired RMSE reduction = 17.5%.

Reliability: the 90% error tolerance

90% absolute-error tolerance (°)

BW	100 mm	200 mm	300 mm	400 mm	500 mm	598 mm
1%	23.1	16.4	12.5	10.5	9.7	**9.4**
5%	21.8	13.9	9.9	8.5	7.7	**7.3**
25%	19.7	12.8	8.4	6.3	5.2	**5.0**
50%	16.9	11.0	6.4	5.4	3.9	**2.8**
75%	13.2	8.6	5.2	4.2	2.6	**1.5**

Available span (mm)

Systematic 50 × 1; pooled empirical absolute errors.
75% BW, 598 mm: 1.5° tolerance, versus 2.5° for random.

5% BW, 598 mm — configuration RMSE and pooled tolerance

Strategy	Median	P90	90% tol.
Random	3.36°	7.95°	8.1°
Systematic	2.37°	7.22°	7.3°

DESIGN GUIDANCE
Cover more space; distribute observations evenly.
Residual phase offsets still set an accuracy limit.

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